

Supplementary information for: Destruction of long-range order in non-collinear two-dimensional antiferromagnets by random-bond disorder

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I. SINGLE BOND DEFECT

Here we provide a few details concerning the physics of a single defect bond.

A. Spin texture via linear response

As explained in the main text, the spin texture induced by a bond defect can be understood as the response to a local dipolar transverse field which acts to rotate the two spins on the defect bond away from their clean-limit configuration. This is most efficiently calculated in a local frame where the clean-limit state is uniform.

To this end we start from a coplanar state in the $x-y$ plane and rotate the spins such that all point along the $\hat{x}$ axis. The rotation involves the y components of the spins, i.e., the susceptibility $\chi^\perp(\vec{q}) = \langle S^y; S^y \rangle$. For a system with underlying SU(2) spin symmetry, this susceptibility is governed by the in-plane Goldstone modes and generically behaves as $\chi^\perp(\vec{q}) \propto 1/q^2$ for small momenta. For the triangular-lattice Heisenberg model in the classical limit an explicit calculation gives

$$\chi^\perp(\vec{q}) = \dots \quad (\text{S1})$$

where we have used ... **[ToDo: fill in]**.

For a bond defect on sites $(0, 1)$ the transverse response at site r is given by

$$\langle S_r^\perp \rangle = \sum_{j=0}^1 h^\perp \beta_j \langle S_r^y; S_j^y \rangle \quad (\text{S2})$$

where $\beta_j = (-1)^j$ is the dipolar form factor of the perturbation, and $h^\perp \propto \delta J$ measures the strength of the perturbation. **[ToDo: write detailed result including direction dependence]**

B. Uniform moment

In the presence of a single defect bond, the bulk spin moments do not exactly cancel each other, such that the resulting state has a non-vanishing uniform magnetization m_{imp} . For small δJ this is proportional to δJ ,

$$m_{\text{imp}} = \gamma \delta J + \mathcal{O}(S^0). \quad (\text{S3})$$

For the triangular-lattice Heisenberg model we numerically find $\gamma =$ **[ToDo: add number]**, see Fig. **[ToDo: add fig]**. The proportionality can be expected to break down for larger δJ , but still holds to very good accuracy for $\delta J = -J$ where we find $m_{\text{imp}} = 0.41S + \mathcal{O}(S^0)$. **[ToDo: can you get the $1/S$ correction?]**

A finite-size scaling analysis of m_{imp} is shown in Fig. **[ToDo: add fig]**.

II. DESTRUCTION OF LONG-RANGE ORDER

As explained in the main text, an estimate of the transverse spin fluctuations induced by a collection of bond defects can be used to assess the destruction of long-range order.

... can be written as ... **[ToDo: check]**

$$\overline{\langle S_t^{\perp 2} \rangle} = 3\alpha(\delta J)^2 \int d^d q (q_x \chi^\perp(\vec{q}))^2 \quad (\text{S4})$$

A. SU(2) symmetry and $T = 0$

For SU(2) symmetry we have $\chi^\perp(\vec{q}) \propto 1/q^2$ and directly obtain ... **[ToDo: fill in]**. This implies a correlation length $\xi^{2-d} \propto (J/\delta J)^2$ for $d < 2$, while in $d = 2$

$$\xi \propto e^{(J/\delta J)^2}. \quad (\text{S5})$$

B. Finite T

At finite T ... thermal correlation length ... **[ToDo: work out crossover]**

C. Layered systems

A realistic layered system displays a small inter-layer coupling which tends to stabilize magnetic order at finite temperatures in the presence of SU(2) symmetry. We assume an (unfrustrated) inter-layer coupling of magnitude $J_\perp$. **[ToDo: write down susceptibility and work out transition criterion]**

D. Broken SU(2)

For a system with broken SU(2) symmetry, some or all of the Goldstone modes acquire a gap. For simplicity we assume

$$\chi^\perp(\vec{q}) = \frac{\chi_0}{q^2 + \Delta^2}. \quad (\text{S6})$$

As a result ... **[ToDo: workout transition criterion]**

III. INCOMMENSURABILITY IN ANISOTROPIC HEISENBERG MODEL

We consider a spatially anisotropic Heisenberg model on the triangular lattice, with couplings $J_{1,2,3}$ on the bonds 1,2,3, with mean J . In the classical limit, the ground state is given by a single- Q coplanar spiral whose propagation wavevector $\vec{Q}$ can be found by minimizing

the Fourier-transformed coupling constant $J(\vec{q})$. In the isotropic case, the system orders at $\vec{Q}_0 = \pm(4\pi/3, 0)$.

The degree of anisotropy can be parameterized by $\vec{d} = \sum_{i=1}^3 (J_i/J) \hat{e}_i$ where $\hat{e}_i$ are the directed unit vectors along the nearest-neighbor bonds of the lattice. As explained in the main text, $\vec{d}$ takes the role of a local dipole density in the bond-disordered case.

For small deviations from isotropy, $|\vec{d}| \ll 1$, the relation between $\vec{Q}$ and $\vec{d}$ can be determined analytically **[ToDo: work out, write down]**.

IV. NUMERICAL RESULTS

V. BOND DEFECT IN THREE DIMENSIONS

We have checked that our scenario for bond disorder applies to three space dimensions as well. To this end, we have considered a single bond defect in **[ToDo: describe model and results]**.